

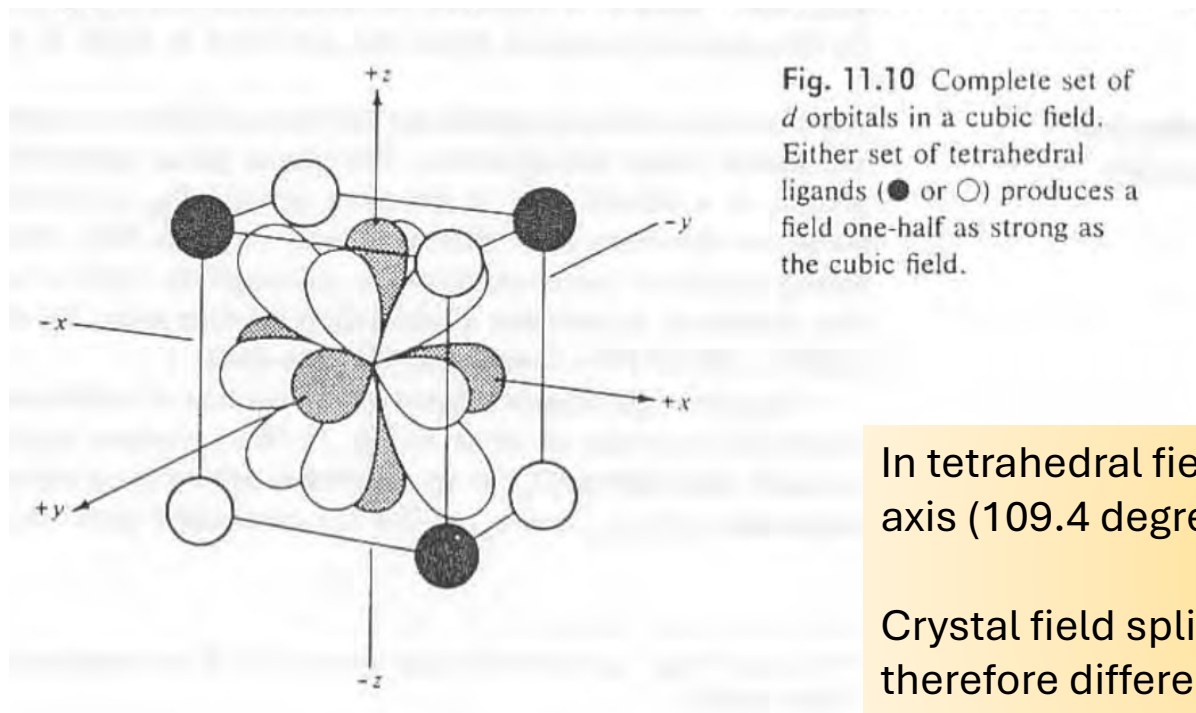
CFT for Tetrahedral geometry

Td will also split d orbitals, but the interactions with ligands are less intense than that in Oh

Stabilization will be less; Destabilization will also be less

Head-on collision between metal and ligand in Octahedral (Oh) geometry
In Tetrahedral, they are just touching

Tetrahedral (Td) field

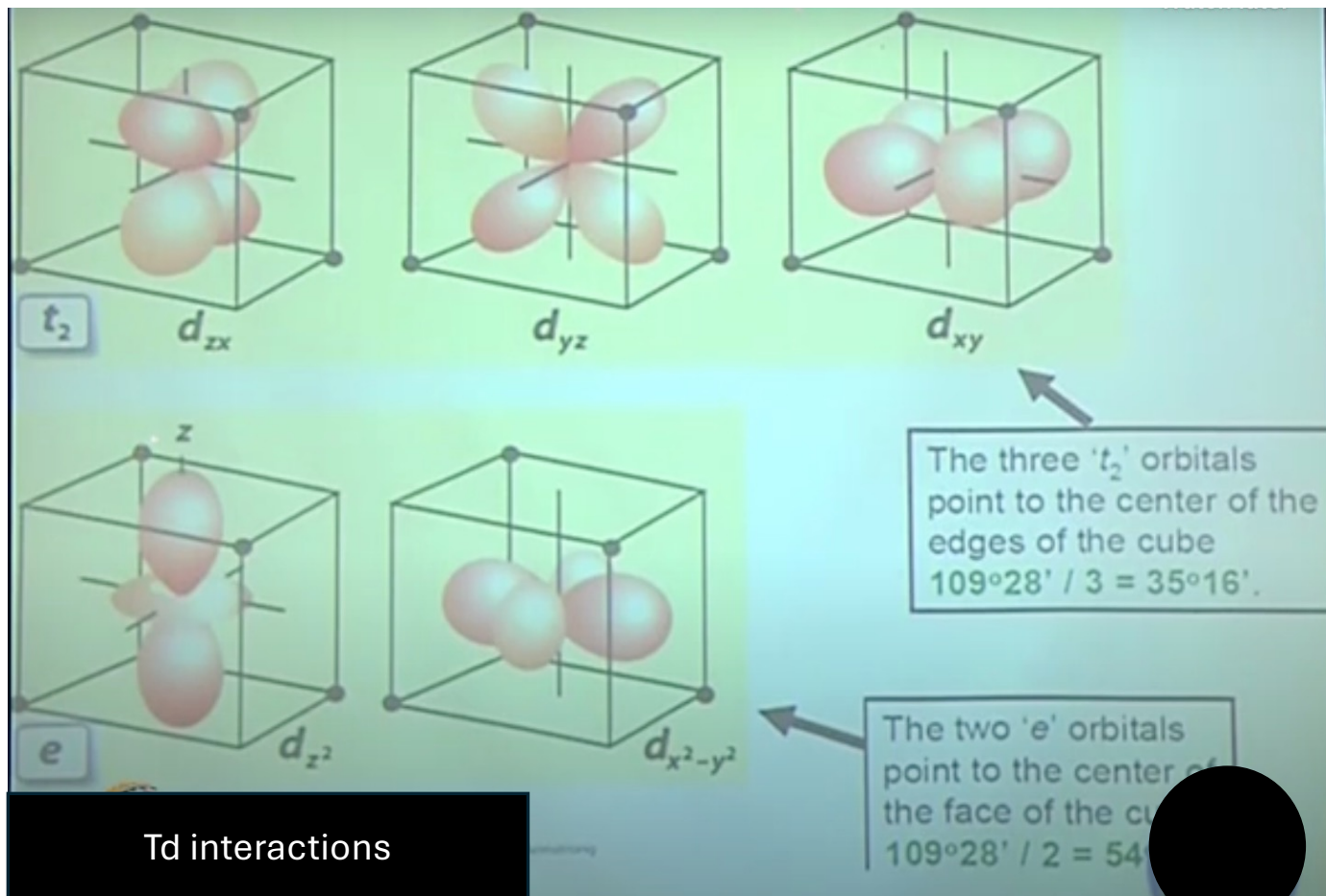


In tetrahedral field, ligands don't approach along the axis (109.4 degree)

Crystal field splitting of d-orbitals in Td geometry are therefore different as compared to Oh geometry

Huheey, Keiter and Keiter

Ligand Interactions in a Tetrahedral (Td) field

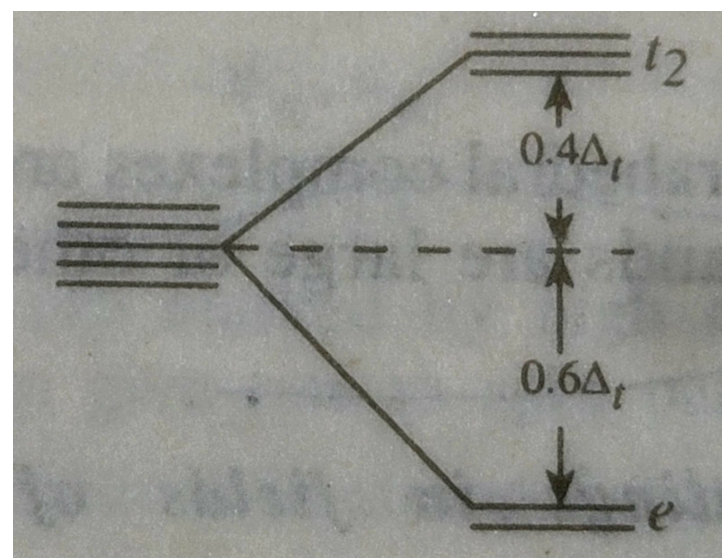
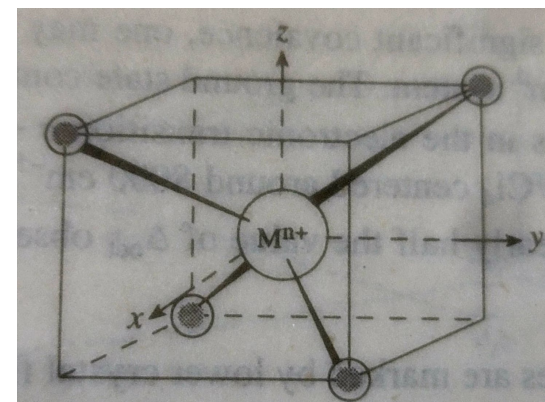


Splitting of d-orbitals in a Tetrahedral (Td) field

$e (d_{x^2-y^2}, d_{z^2})$ - stabilized

$t_2 (d_{xy}, d_{yz}, d_{zx})$ - destabilized

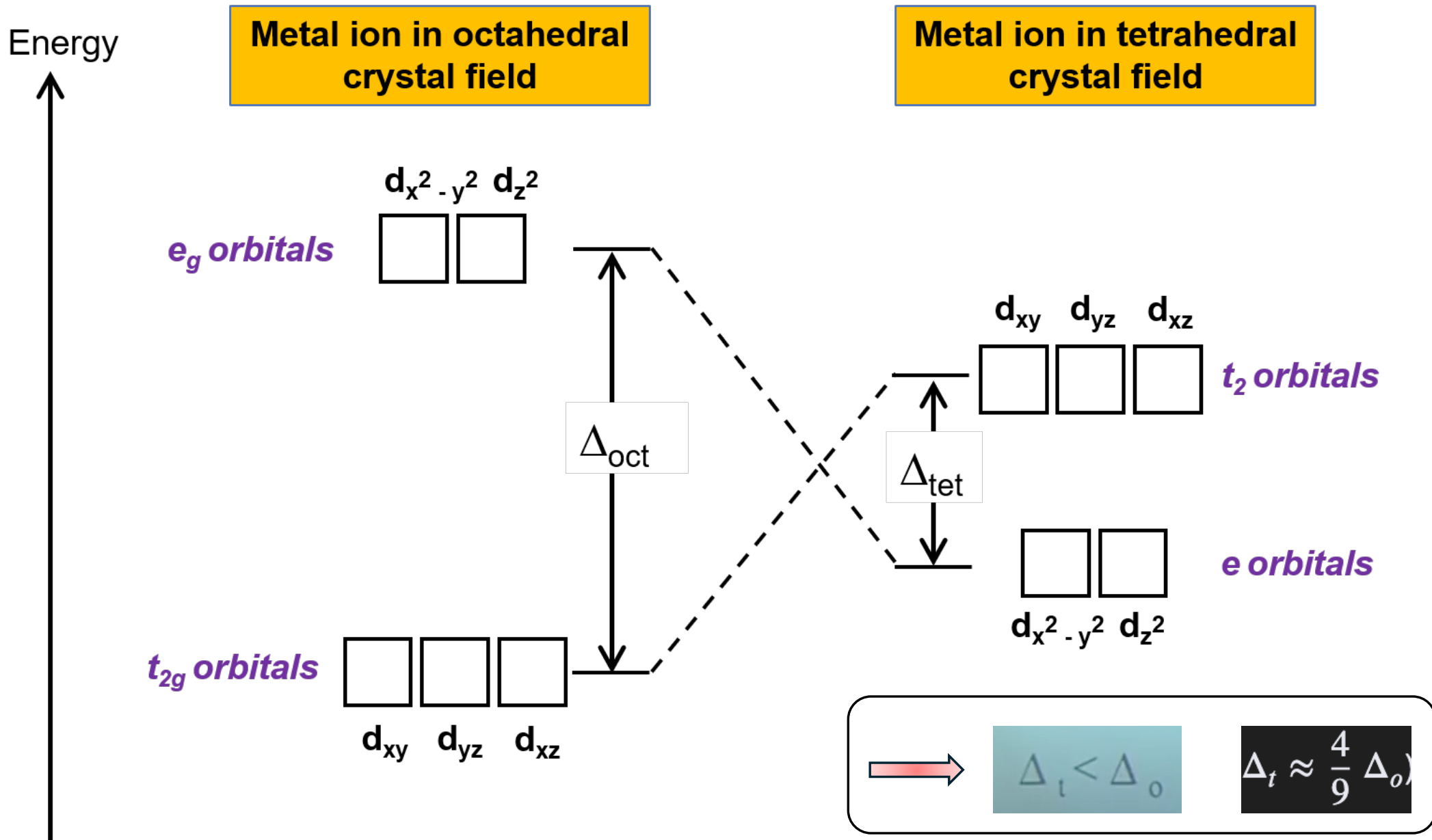
No “g”



g stands for “gerade” (German: even)

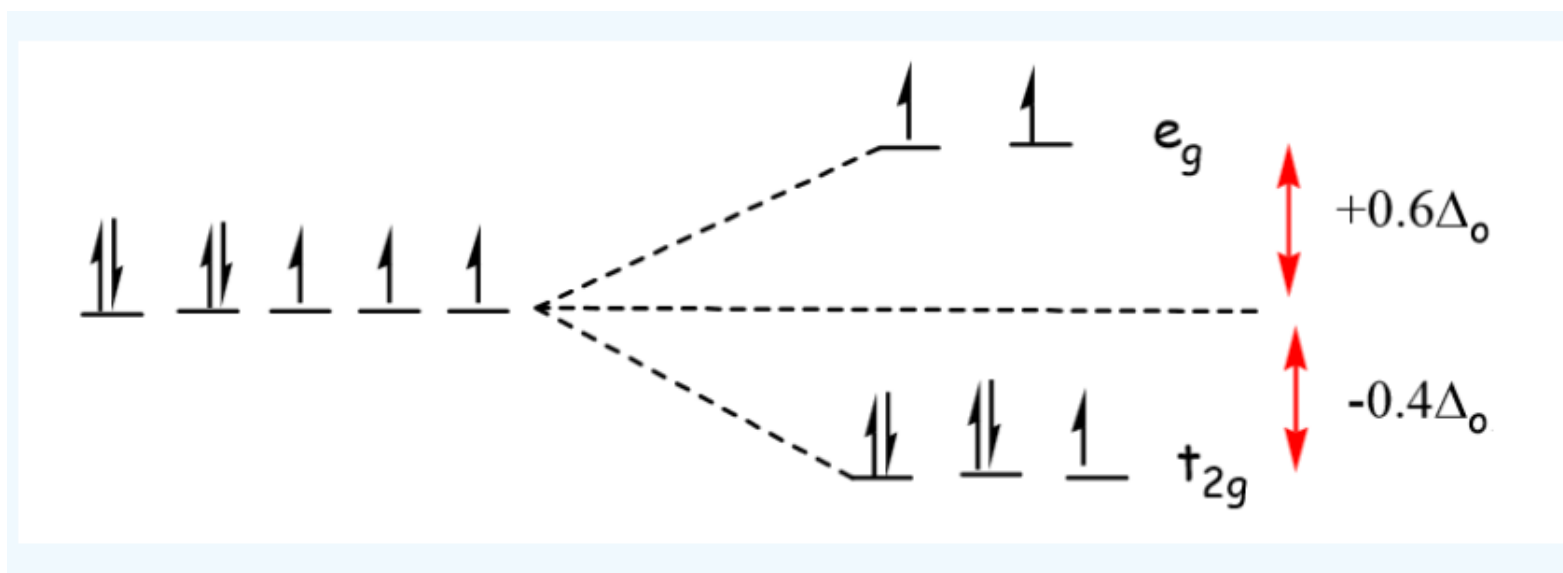
Basically, signifies that inversion center (symmetry) is present in Oh but absent in Td

Oh is symmetrical around origin, but Td is not



Crystal Field Stabilization Energy (CFSE)

Energy diagram of d7 in a non-symmetrical field where all d orbitals lose degeneracy



How to calculate CFSE for complexes?

$$= 5 \times (-0.4) + (2 \times 0.6) \Delta_o$$

$$= -2.0 + 1.2 \Delta_o = -0.8 \Delta_o$$

CFSE for d^1 system?

CFSE for d^2 system?

CFSE for d^3 system?

Octahedral Field

$$\text{For } d^1, \text{ CFSE} = 1 \times -0.4 = -0.4 \Delta_o$$

$$\text{For } d^2, \text{ CFSE} = 2 \times -0.4 = -0.8 \Delta_o$$

$$\text{For } d^3, \text{ CFSE} = 3 \times -0.4 = -1.2 \Delta_o$$

Calculate CFSE in Td field for d^1 , d^2 , d^3 systems?

In tetrahedral field:

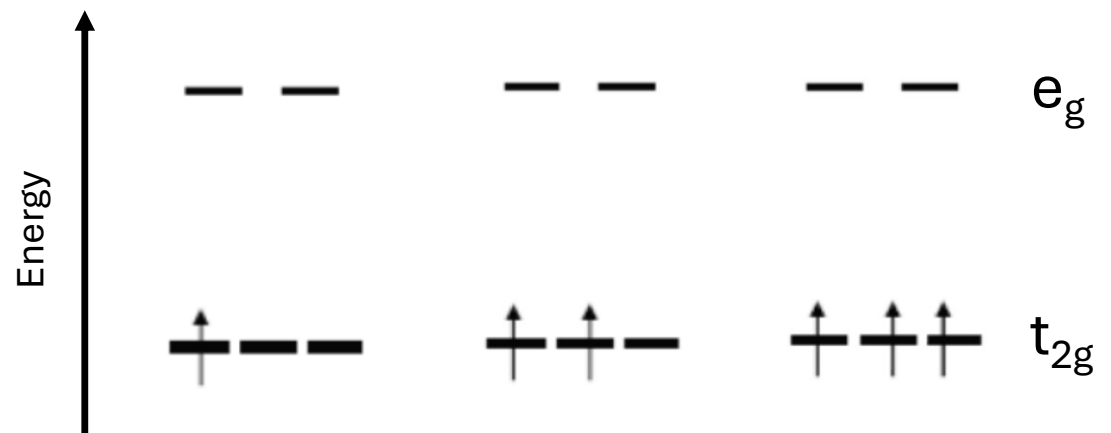
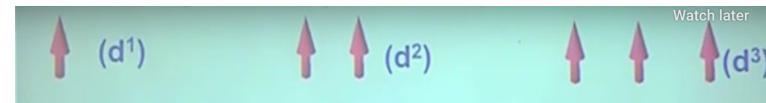
e is ground state and t_2 is excited state

CFSE will be in Δ_t

How to fill up electrons in CFT orbitals

- **Hund's rule** of maximum multiplicity (number of electrons having parallel spin)
- **Pauli's exclusion principle** (opposite spin)
- Crystal field splitting energy
- Pairing energy (P.E.)

- For d^1 - d^3 systems: t_{2g}^3 (Hund's rule of maximum multiplicity)

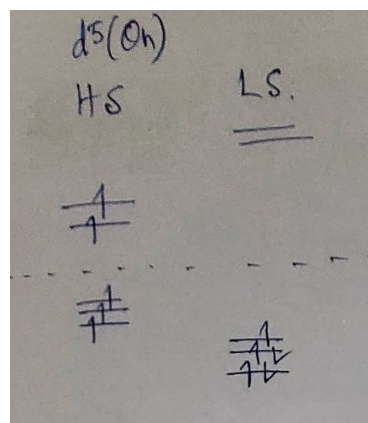


How to fill up electrons in CFT orbitals

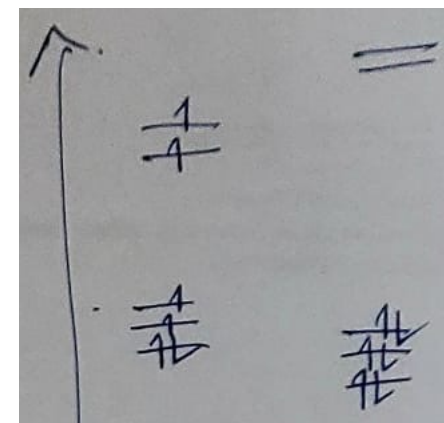
- **Hund's rule** of maximum multiplicity (number of electrons having parallel spin)
- **Pauli's exclusion principle** (opposite spin)
- Crystal field splitting energy
- Pairing energy (P.E.)

The concept of High spin (HS) and Low spin (LS) complexes

d5 Octahedral



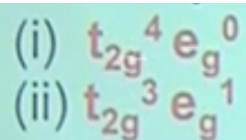
d6 Octahedral



What governs spin state?
How do we calculate the CFSE?

The concept of High Spin and Low Spin Complexes

For d^4 - d^7 systems:



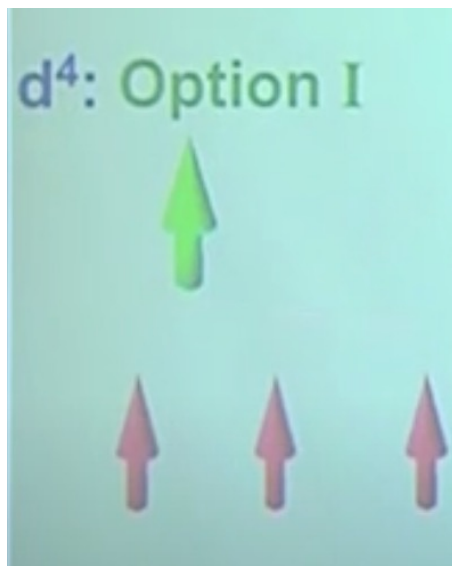
(low spin case or strong field situation)
(high spin case or weak field situation)

Parameters (for HS and LS case):

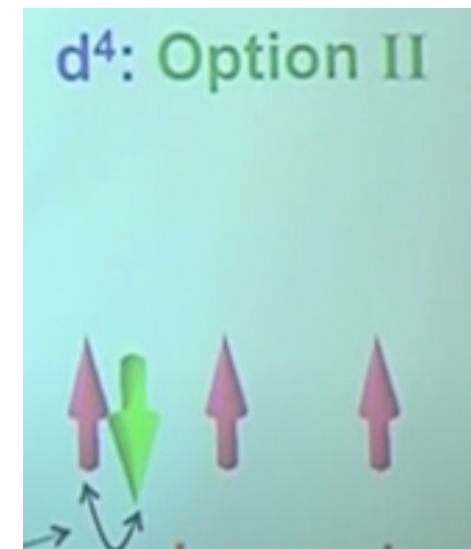
- (i) **CFSE** (value of Δ_0 , $10Dq$)
- (ii) **Pairing energy** of e^- (repulsive)

- $CFSE > P.E.$: **LS** complex
- $CFSE < P.E.$: **HS** complex

Low spin or High spin: Pairing Energy vs Splitting (Δ_o)



So which one?
Decided by
(i) Δ_o
(ii) Pairing E.



Factors affecting the splitting (Δ_o)

- the nature of the metal ion.
- the metal's oxidation state. A higher oxidation state leads to a larger splitting relative to the spherical field.
- the arrangement of the ligands around the metal ion.
- the coordination number of the metal (i.e. tetrahedral, octahedral...)
- the nature of the ligands surrounding the metal ion. The stronger the effect of the ligands then the greater the difference between the high and low energy d groups.

- *Nature of the ligands*
- *The **charge** on the metal ion*
- *Whether the metal is a **3d**, **4d**, or **5d** element*

Ligands: *Weak field ligands* ; small splitting ($\Delta_o \sim 7000 - 30000 \text{ cm}^{-1}$)

Strong field ligands; large splitting ($\Delta_o > 30000 \text{ cm}^{-1}$)

Knowledge of the **charge** and **denticity** of ligands are important for deducing charge of metal and coordination geometry

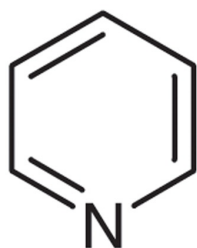
Negatively charged ligands

Cl⁻, F⁻, Br⁻, I⁻, OH⁻, CN⁻, etc

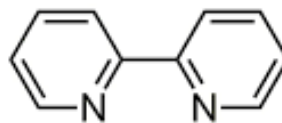
Neutral ligands

CO, PPh₃, H₂O, NH₃

Aromatic neutral ligands

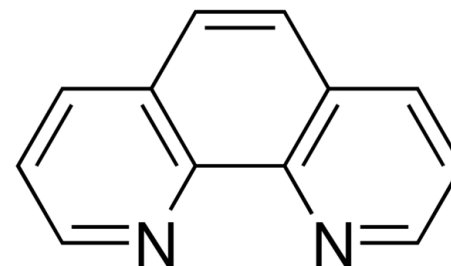


pyridine



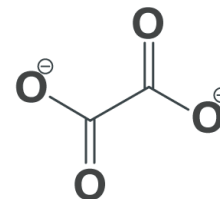
bpy

bipyridine



phen (phenanthroline)

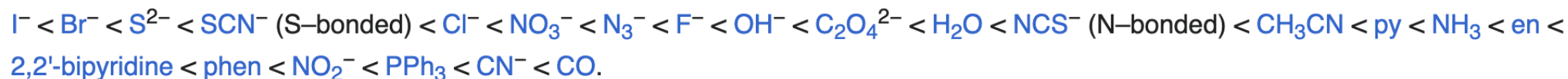
Negatively charged bidentate ligand



oxalate

Spectrochemical Series

The spectrochemical series is an empirically-derived list of ligands ordered by the size of the splitting Δ that they produce



Increasing Δ_o

Spectrochemical series for metal ions



strong-field ligands

Decreasing Δ_o

weak-field ligands

Parameters deciding HS and LS configurations

Complex	Config.	Δ_o , cm^{-1}	P, cm^{-1}	spin-state
$[\text{Fe}(\text{OH}_2)_6]^{2+}$	d^6	10,400	17,600	high-spin
$[\text{Fe}(\text{CN})_6]^{4-}$	d^6	32,850	17,600	low-spin

P = pairing energy

Low spin complexes are characterized by
High oxidation state of metal
Strong ligands

High spin complexes are characterized by
Low oxidation state of metal
Weak ligands

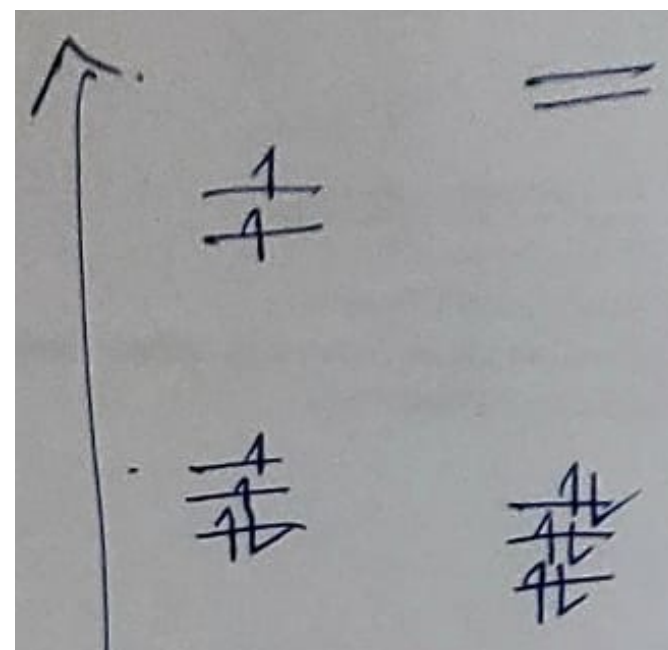
CFT also explains magnetic behaviour

$[\text{Co}(\text{F})_6]^{3+}$ is paramagnetic, but
 $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic.
Explain using CFT.

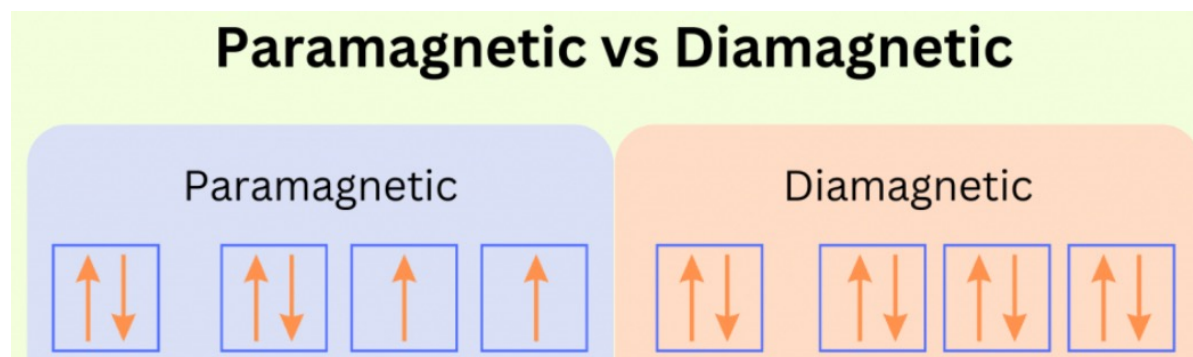
Co^{3+} and Fe^{2+} are both d6 systems
Coordination number = 6, therefore Oh geometry
t2g, eg

F is weak ligand – so high spin complex – t2g4eg2
CN is strong ligand – so low spin complex – t2g6eg0

d6 Octahedral



$[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic, but
 $[\text{Fe}(\text{H}_2\text{O})_6]^{4+}$ is diamagnetic.
Explain using CFT.



Unpaired electrons:

YES

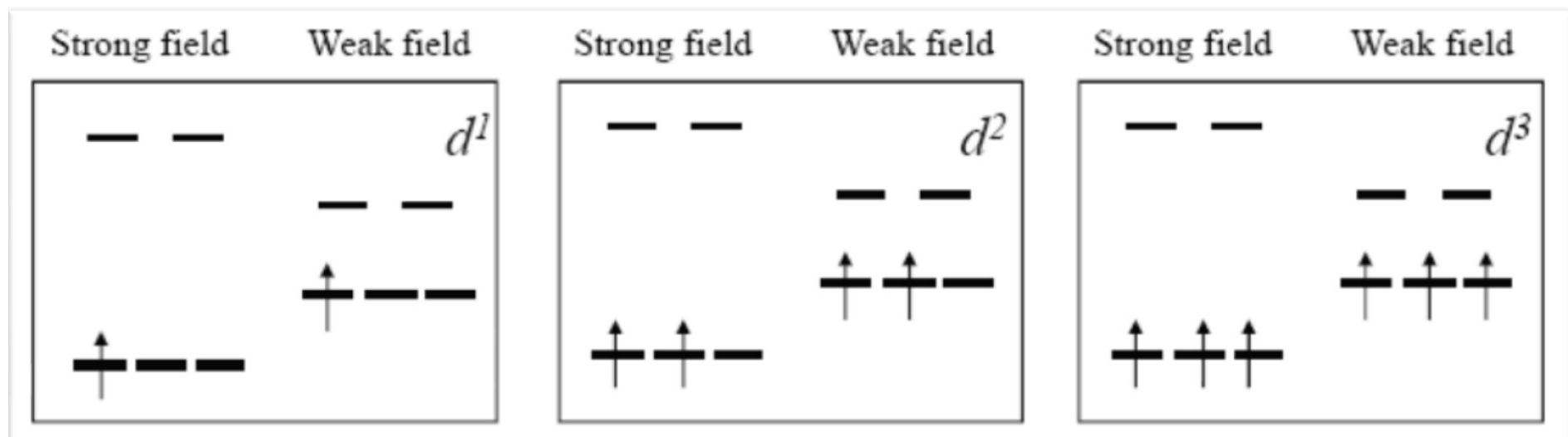
NO

CFSE: Summary of possible d configurations in strong and weak field

d^n	Weak field			Strong field		
	Configuration	Unpaired electrons	CFSE	Configuration	Unpaired electrons	CFSE
d^1	t_{2g}^1	1	$0.4\Delta_o$	t_{2g}^1	1	$0.4\Delta_o$
d^2	t_{2g}^2	2	$0.8\Delta_o$	t_{2g}^2	2	$0.8\Delta_o$
d^3	t_{2g}^3	3	$1.2\Delta_o$	t_{2g}^3	3	$1.2\Delta_o$
d^4	$t_{2g}^3 e_g^1$	4	$0.6\Delta_o$	t_{2g}^4	2	$1.6\Delta_o$
d^5	$t_{2g}^3 e_g^2$	5	$0.0\Delta_o$	t_{2g}^5	1	$2.0\Delta_o$
d^6	$t_{2g}^4 e_g^2$	4	$0.4\Delta_o$	t_{2g}^6	0	$2.4\Delta_o$
d^7	$t_{2g}^5 e_g^2$	3	$0.8\Delta_o$	$t_{2g}^6 e_g^1$	1	$1.8\Delta_o$
d^8	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$	$t_{2g}^6 e_g^2$	2	$1.2\Delta_o$
d^9	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$	$t_{2g}^6 e_g^3$	1	$0.6\Delta_o$
d^{10}	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$	$t_{2g}^6 e_g^4$	0	$0.0\Delta_o$

^a This table is somewhat simplified because pairing energies and electron-electron effects have been neglected.

Draw the CFSE diagram and show the calculation



Crystal field splitting is higher in strong field ligands

In Td complexes, low spin complexes are rarely observed, because splitting (Δ) is always lower than P.E.

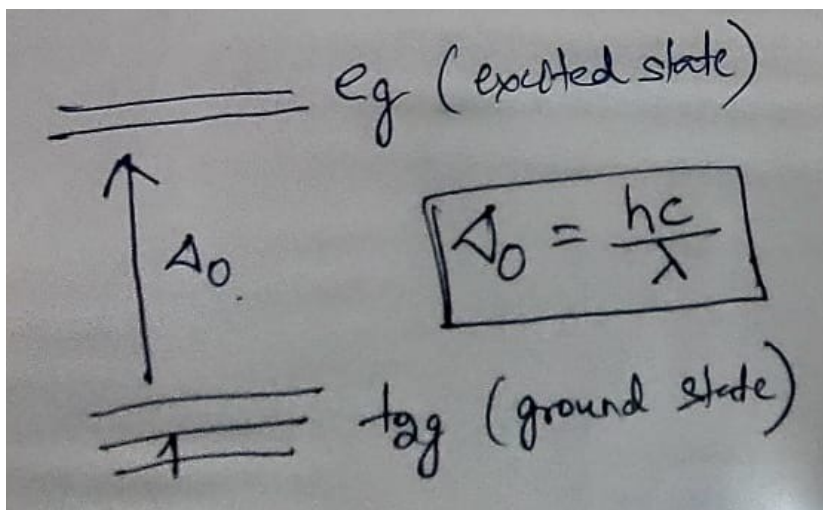
Δ_0 values depend on Ligand and Metal

wavenumber

$$\bar{\nu} = \frac{1}{\lambda}$$

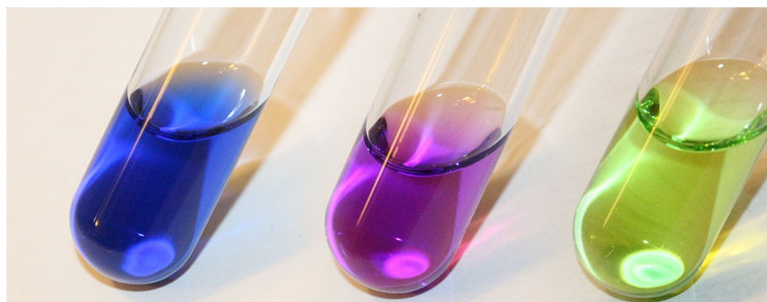
Δ_0

$[\text{CrCl}_6]^{3-}$	13640 cm^{-1}	163 kJ/mol
$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	17830	213
$[\text{Cr}(\text{NH}_3)_6]^{3+}$	21680	314
$[\text{Cr}(\text{CN})_6]^{3-}$	26280	314
$[\text{Co}(\text{NH}_3)_6]^{3+}$	24800 cm^{-1}	163 kJ/mol
$[\text{Rh}(\text{NH}_3)_6]^{3+}$	34000	213
$[\text{Ir}(\text{NH}_3)_6]^{3+}$	41000	314
$3d < 4d < 5d$ $M^{2+} < M^{3+} < M^{4+}$		



Splitting will depend on energy gap

Therefore, different metals, different oxidation states, different geometry, different ligands will affect the wavelength of transition



This explains why colors of different complexes are different.

But why different intensities?

Intensity directly correlates with absorption coefficient

Beer-Lambert Law

$$A = \epsilon c \ell,$$

where

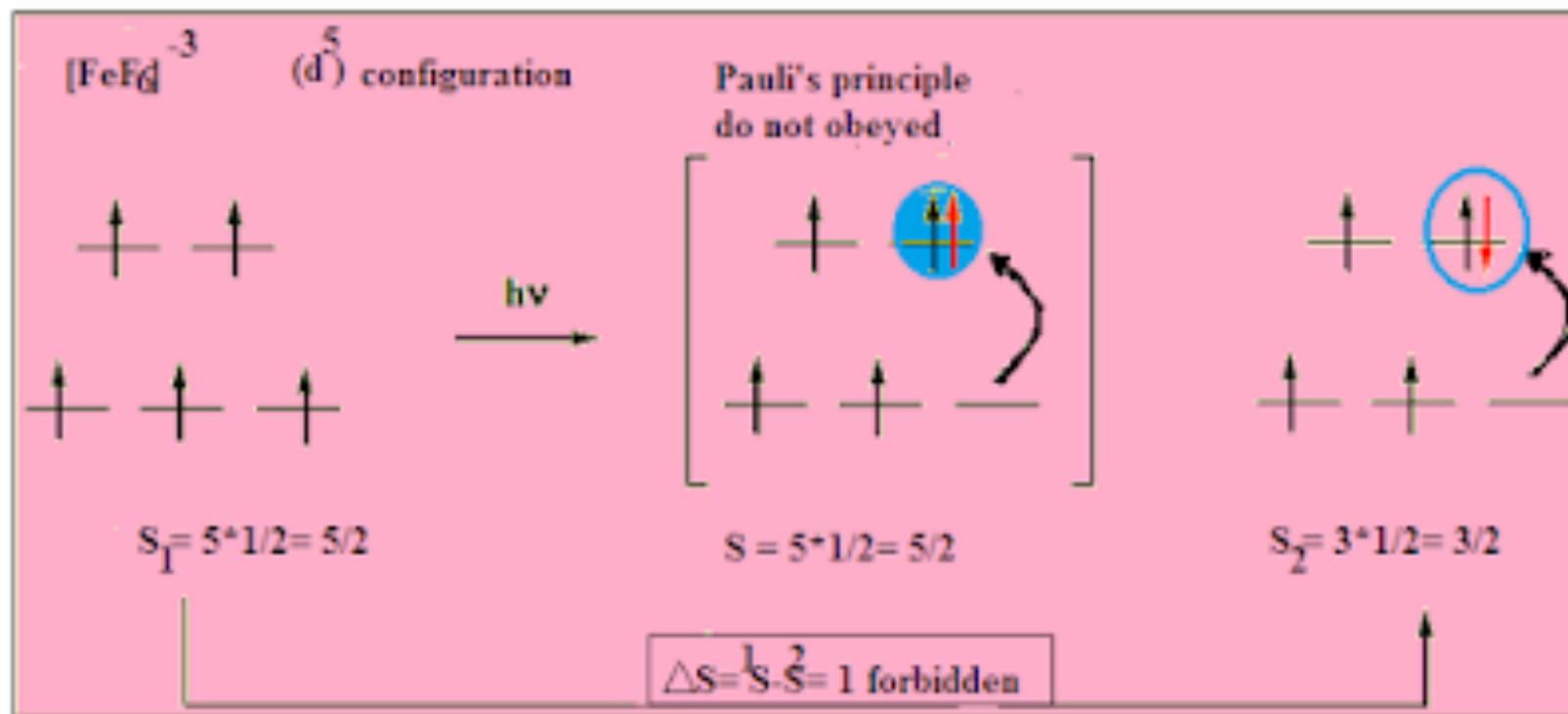
- ϵ is the *molar absorption coefficient* of that material;
- c is the **molar concentration** of those species;
- ℓ is the path length.

Extinction coefficient → **probability of an electronic transition**

Probabilities of transition are mainly decided by a two rules

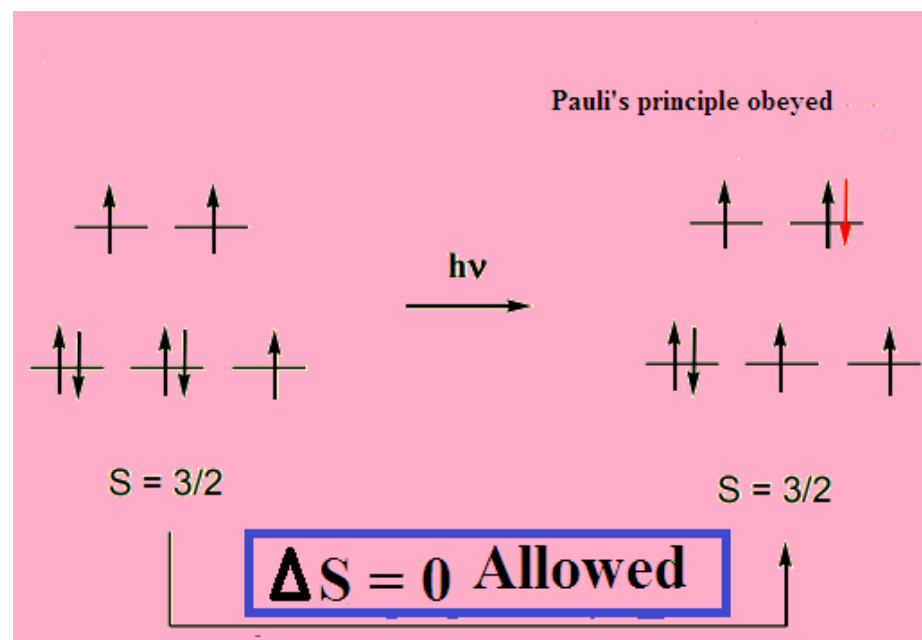
1. **The Spin Selection Rule, $\Delta S = 0$.**

A spin-forbidden transition is an electronic jump between states with different total spin quantum numbers, violating the rule that $\Delta S = 0$ (spin multiplicity must stay the same) for an allowed transition, making it less probable but not impossible, often observed as weak spectral lines due to **spin-orbit coupling mixing states**



d⁵ configuration in octahedral geometry results in extremely weak absorption and low extinction coefficient. Metal complexes show faint color

Transition in d7 High Spin system



The Laporte selection rule dictates that electronic transitions between orbitals of the **same parity** ($g \rightarrow g$ or $u \rightarrow u$) are forbidden in centrosymmetric molecules. Primarily affecting octahedral and square-planar complexes, it forbids transitions like $s \rightarrow s$, $d \rightarrow d$, and $p \rightarrow f$. These transitions often appear as weak absorption bands because they are partially allowed through **vibronic coupling**.

$g \rightarrow u$ or $u \rightarrow g$ are allowed transitions

📌 Breaking the rules!

Relaxation of the Laporte and Spin Selection Rules rules can occur through:

- **Vibronic coupling:** the bonds of metal complexes vibrate and may cause temporary distortions in molecular symmetry. These distortions can cause temporary loss in symmetry (as well as some orbital mixing), and thus allow $d \rightarrow d$ transitions in those moments of distortion. Despite being forbidden by the Laporte Selection rule, the $d \rightarrow d$ transitions in octahedral complexes appear, but are weak (of low intensity) with molar absorptivities $\leq 100 \text{ M}^{-1}\text{cm}^{-1}$.
- **Orbital Mixing:** In the case of octahedral complexes, π -acceptor and π -donor ligands can mix with the d-orbitals so that transitions are no longer purely $d \rightarrow d$ (but these are usually considered "charge transfer" transitions, not $d \rightarrow d$). In the case of tetrahedral complexes, the molecule has no center of symmetry (thus no u or g subscripts on the terms). In the valence orbital model, a tetrahedral molecule is said to have sp^3 and sd^3 hybridized orbitals, while MO theory predicts MO's with mixtures of some s, p, and d character - this mixing of orbital types can allow transitions between "mixed" orbitals that would otherwise be forbidden in "pure" orbitals. A similar phenomenon would occur in octahedrons with significant distortion, or in other coordination geometries that provide potential for orbital mixing.
- **Spin-Orbit coupling:** This gives rise to spin-forbidden bands of low intensity, usually with very weak molar absorptivities approximately $\leq 1 \text{ M}^{-1}\text{cm}^{-1}$. This phenomenon is usually more important for transition metals of the second row $4d$ and beyond.

d-d transitions in octahedral geometry are Laporte forbidden – weak transitions – less intense bands

d-d transitions in octahedral geometry may or may not be spin forbidden – if forbidden, then weak absorption – less intense bands

d-d transitions in tetrahedral geometry are not Laporte forbidden – strong absorbance, more intense bands

d-d transitions in tetrahedral geometry may or may not be spin forbidden